Moles and Molar Quantities Unit: Sample Test, TEACHER KEY

Every value is computed in code (atomic masses to 0.01 g/mol, NA = 6.022×10^{23}, 22.4 L/mol at STP) and rounded to the sig figs of the data. Accept answers within one rounding digit if the work is correct. Points are listed in each part.

Part I: Multiple Choice (1 point each)

#	Ans	Correct choice	Why
1	**B**	One mole is 6.022×10^{23} particles.	The mole counts particles (atoms, molecules or formula units). A compound's molecules each hold several atoms, and 22.4 L is only for a gas at STP.
2	**D**	1.51×10^{23} atoms	0.250 mol × 6.022×10^{23} atoms/mol
3	**A**	164.10 g/mol	40.08 + 2(14.01) + 6(16.00): the 2 outside the parentheses multiplies everything inside.
4	**C**	0.5000 mol	$3.011 \times 10^{23} \div 6.022 \times 10^{23}$ = 0.5000 mol
5	**B**	146 g	2.50 mol × 58.44 g/mol = 146 g
6	**A**	2.00 mol	88.0 g ÷ 44.01 g/mol = 2.00 mol
7	**D**	1.0 mol of CH_4	Every sample is 1.0 mol of particles, but a CH_4 molecule holds 5 atoms (H_2 and N_2 hold 2, He holds 1).
8	**B**	Divide by the molar mass.	Grams and particles never convert directly: grams → moles (÷ molar mass) → particles (× 6.022×10^{23}).
9	**C**	11.2 L	0.500 mol × 22.4 L/mol = 11.2 L
10	**D**	1.50 mol	33.6 L ÷ 22.4 L/mol = 1.50 mol
11	**A**	H_2	Volume follows moles, and the lightest gas has the most moles in 10.0 g: H_2 4.95 mol, He 2.50, N_2 0.357, CO_2 0.227.
12	**C**	88.79%	16.00 g ÷ 18.02 g × 100 = 88.79%
13	**A**	35.00%	Two N atoms: 28.02 g ÷ 80.06 g × 100 = 35.00%. The formula has two nitrogens (one in NH_4^+, one in NO_3^-).
14	**B**	CH_2O	Divide every subscript by 6 (the largest whole number that divides them all).
15	**C**	CH_4	75.0 g C = 6.245 mol; 25.0 g H = 24.75 mol; ratio C : H = 1 : 3.96, which is 1 : 4.
16	**D**	$C_2H_4O_2$	Empirical mass = 30.03 g/mol; 60.06 ÷ 30.03 = 2, so multiply every subscript by 2.
17	**A**	copper(II) sulfate pentahydrate	Copper(II) because sulfate is 2−; the prefix penta- gives the 5 water molecules.
18	**B**	36.0%	Water lost = 5.00 − 3.20 = 1.80 g; 1.80 ÷ 5.00 × 100 = 36.0%.
19	**C**	5	0.0498 mol H_2O ÷ 0.0100 mol $CuSO_4$ = 4.98, which rounds to 5.
20	**D**	Each Ne atom has a greater mass than each He atom.	A mole is a count (6.022×10^{23}). Ne is 20.18 g/mol and He is 4.00 g/mol because each Ne atom is heavier.

Answer pattern: 1B 2D 3A 4C 5B 6A 7D 8B 9C 10D 11A 12C 13A 14B 15C 16D 17A 18B 19C 20D

Part II: Free Response (46 points)

1) Calculate each quantity. Show your setup with units and give the answer with the correct sig figs. (12 pts)

a. Find the molar mass of $(NH_4)_2CO_3$. (2 pts)

N: 2 × 14.01 = 28.02; H: 8 × 1.01 = 8.08; C: 12.01; O: 3 × 16.00 = 48.00

Molar mass = **96.11 g/mol**

b. How many grams are in 0.750 mol of $Mg(OH)_2$? (2 pts)

$Mg(OH)_2$ = 24.31 + 2(16.00) + 2(1.01) = 58.33 g/mol

0.750 mol × 58.33 g/mol = **43.7 g**

c. How many molecules are in 25.0 g of glucose, $C_6H_{12}O_6$? (3 pts)

$C_6H_{12}O_6$ = 180.18 g/mol

25.0 g × (1 mol / 180.18 g) × (6.022×10^{23} molecules / 1 mol) = **8.36×10^{22} molecules**

d. How many oxygen atoms are in 0.400 mol of $Al_2(SO_4)_3$? (3 pts)

Each formula unit has 12 O atoms (4 × 3).

0.400 mol $Al_2(SO_4)_3$ × (12 mol O / 1 mol) × (6.022×10^{23} atoms / 1 mol) = **2.89×10^{24} O atoms**

e. What is the mass of 3.011×10^{23} molecules of NH_3? (2 pts)

$3.011 \times 10^{23} \div 6.022 \times 10^{23}$ = 0.5000 mol; NH_3 = 17.04 g/mol

0.5000 mol × 17.04 g/mol = **8.520 g**

2) Gases at STP. Use 22.4 L/mol where it applies. (10 pts)

a. What volume does 14.2 g of Cl_2 gas occupy at STP? (3 pts)

Cl_2 = 70.90 g/mol

14.2 g × (1 mol / 70.90 g) × (22.4 L / 1 mol) = **4.49 L**

b. What is the mass of 5.60 L of NH_3 gas at STP? (3 pts)

5.60 L × (1 mol / 22.4 L) × (17.04 g / 1 mol) = **4.26 g**

c. How many molecules are in 11.2 L of N_2 gas at STP? (2 pts)
11.2 L × (1 mol / 22.4 L) × (6.022×10^{23} molecules / 1 mol) = **3.01×10^{23} molecules**

d. A student uses 22.4 L/mol for a gas at 25 °C and 1 atm. Explain why the answer will be wrong. (2 pts)
22.4 L/mol is the molar volume only at STP (0 °C and 1 atm). At 25 °C the gas is warmer, so 1 mol takes up more than 22.4 L.

3) Percent composition and formulas. (12 pts)

a. Find the percent by mass of each element in $Al_2(SO_4)_3$. (4 pts)
Molar mass = 2(26.98) + 3(32.06) + 12(16.00) = 342.14 g/mol
Al: 53.96 ÷ 342.14 × 100 = **15.77%**; S: 96.18 ÷ 342.14 × 100 = **28.11%**; O: 192.00 ÷ 342.14 × 100 = **56.12%**
(check: the percents add to 100%)

b. A compound is 40.00% C, 6.71% H and 53.29% O by mass and has a molar mass of 180 g/mol. Find its empirical formula and its molecular formula. Show your work. (8 pts)
Assume 100 g: 40.00 g C, 6.71 g H, 53.29 g O (2 pts)
Moles: C 40.00 ÷ 12.01 = 3.331; H 6.71 ÷ 1.01 = 6.644; O 53.29 ÷ 16.00 = 3.331 (2 pts)
Divide by the smallest (3.331): C 1.000, H 1.99, O 1.000 → **empirical formula CH_2O** (2 pts)
Empirical mass = 12.01 + 2(1.01) + 16.00 = 30.03 g/mol; 180 ÷ 30.03 = 5.99, so n = 6 → **molecular formula $C_6H_{12}O_6$** (2 pts)

4) Hydrates. A student heats a sample of blue copper(II) sulfate hydrate, $CuSO_4 \cdot xH_2O$, in a crucible until the mass stops changing. (12 pts)

Data: mass of empty crucible = 18.42 g; crucible + hydrate (before heating) = 20.92 g; crucible + solid (after heating) = 20.02 g

a. Find the mass of the hydrate, the mass of water lost and the mass of anhydrous $CuSO_4$. (2 pts)
Hydrate: 20.92 − 18.42 = **2.50 g**; water lost: 20.92 − 20.02 = **0.90 g**; anhydrous $CuSO_4$: 20.02 − 18.42 = **1.60 g**

b. Calculate the moles of water lost and the moles of $CuSO_4$ left. (3 pts)
H_2O: 0.90 g ÷ 18.02 g/mol = **0.0499 mol**
$CuSO_4$ (159.61 g/mol): 1.60 g ÷ 159.61 g/mol = **0.0100 mol**

c. Find x. Then write the formula and the name of the hydrate. (3 pts)
x = 0.0499 ÷ 0.0100 = 4.98, which rounds to 5
$CuSO_4 \cdot 5H_2O$, named **copper(II) sulfate pentahydrate**

d. Find the percent water by mass in this hydrate. (2 pts)
0.90 g ÷ 2.50 g × 100 = **36.0%**

e. The student stopped heating early, so some water stayed in the crucible. Will the calculated x be too high or too low? Explain. (2 pts)
Too low. The mass of water lost is too small, so moles of water and the ratio to $CuSO_4$ come out smaller than the true value.